Sentiment Analysis for Streams of Web Data: A Case Study of Brazilian Financial Markets

Bruna Neuenschwander
UFMG, Belo Horizonte/Brasil
bruna@dcc.ufmg.br

Adriano Pereira
UFMG, Belo Horizonte/Brasil
adrianoc@dcc.ufmg.br

Wagner Meira Jr.
UFMG, Belo Horizonte/Brasil
meira@dcc.ufmg.br

Denilson Barbosa
University of Alberta, Edmonton/Canada
denilson@ualberta.ca

ABSTRACT

With the rise of Web 2.0 applications, most people started consuming information and sharing opinions and ideas about most aspects of their lives on a variety of social media platforms, creating massive and continuous streams of valuable data. While this opened the door for information extraction and mining techniques that can help us understand different aspects of society, extracting useful information from such streams of Web data is far from trivial. In this setting, sentiment analysis techniques can be convenient as they are capable of summarizing general feeling about entities people care about, such as products and companies. Therefore, they can be quite applicable in scenarios like the stock market, which also has tremendous impact on society. This paper describes and evaluates two different techniques for sentiment analysis applied to the Brazilian stock market data: lexicon-based and machine learning based, considering a wide range of text pre-processing and feature selection approaches.

Keywords

Sentiment Analysis; Social Data; Machine Learning; Web and Social Networks; Lexicon; Brazilian Stock Market

1. INTRODUCTION

Web 2.0 offers a wide diversity of new communication media, varying from news and *blogs* to social media and discussion forums. This allows users not only to express theirs opinions but also to follow different perspectives and contents. In this environment, users are exposed to vast amounts of information, continuously updated in a streaming fashion, even from people they are not related to. Manually tracking information from a great variety of sources has quickly become unfeasible. Consequently, finding a way to optimize user time and knowledge acquired is valuable. In this sense, data mining techniques that work on text can be extremely useful, helping, for instance, to summarize contents, identify trends and opinions, and ultimately helping predict events [11].

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
WebMedia'14, November 18–21, 2014, João Pessoa, Brazil.
Copyright 2014 ACM 978-1-4503-3230-9/14/11 $15.00.
http://dx.doi.org/10.1145/2664551.2664579.

One of the most well-known areas of textual data mining is called sentiment analysis. At its core, this problem can be defined as labeling a set of texts as positive or negative. There are two main approaches that can be used for sentiment analysis: machine learning based and lexicon-based. Another possibility is to combine both techniques. This combination could lead to a classifier that is less dependent on context, yielding better precision and recall. It is also important to highlight that for texts written in English there are many different resources available. However, for other languages, such as Brazilian Portuguese, the resources are considerably more scarce.

Sentiment analysis applications vary from identifying society mood regarding brands and products [22] to identifying users political views [19]. This kind of analysis can be extremely useful in contexts like stock market. Stocks markets all around the world are responsible for considerable financial transactions, for instance, the Brazilian stock market was one of the top 20 with higher capitalization value in 2011 [2, 3] and had an average value of daily transactions in 2013 of R$ 7.4 billion [1]. Previous studies showed that the behavior of the market can be modified by news [6] and society's mood measured using Twitter data can predict its fluctuations [5]. Therefore, gathering and analyzing information from different sources can provide not only an overview of specialists and investors expectations but can identify new trends (i.e. fluctuation in the frequency of different terms). In this manner, this work shows how different classifiers (lexicon-based and machine learning based) and different pre-processing techniques can improve sentiment analysis for Brazilian stock market data sets.

2. RELATED WORK

Researchers and investors always have the ambition of better understanding the stock market and many studies try to unveil the forces and patterns that supposedly govern it. Two opposing theoretical lines about stock prices prediction are shown in [8]: the chartist and the random walk. The first proposes that stock prices have patterns that repeat over the time and the second states that prices behave randomly. Stock market behavior can also be affected by different factors, such as consumption, and [7] presents an evaluation of these factors.

The Web 2.0 provides different information sources, that along with news, report the latest indexes, companies accomplishments and losses as well as countries economic situation and general opinions. Those sources of information, not only convey the newest facts and expectations but also

could interfere with the behavior of markets. According to [6], investors often do not present a rational response in face of unexpected news, and this overreaction affects the stock market, which can also be seen as motivation for our work. Another interesting discussion in [5] shows that Twitter mood may reflect society's mood and predict stock market changes. Both [5] and [6], reinforce the importance of analyzing data from news and social media and, thereafter, are motivations for applying data mining techniques in this scenario, specially, sentiment analysis.

There is a great variety of publications regarding sentiment analysis such as the survey [16], which is an extensive compilation of applications and techniques for sentiment analysis and opinion mining. On the other hand, [9] is a short review about it and states that there are two main approaches for dealing with sentiment analysis: supervised and unsupervised, both approaches are used in this work.

Among the unsupervised approaches, SOCAL [21] is a prominent lexicon-based method which has been adapted for Brazilian Portuguese [15]. As for the supervised approaches, there is a wide number of machine learning classifiers. In this work, Naive-Bayes classifiers are used along with different pre-processing techniques. Consequently, [17] is very interesting as it shows the performance of three classifiers, including Naive-Bayes, along with pre-processing techniques, which is similar to our work. Furthermore, it shows other possibilities of supervised algorithms and pre-processing techniques that were not explored and could be as future work.

Considering sentiment analysis for Brazilian Portuguese data sets, [20] examines the use of lexicons and negation models for Twitter analysis as well as different heuristics to normalize the texts, for instance, a heuristic to remove repeated vowels. In the context of news, specially in the financial market, [13] describes the construction of a lexicon for financial scenario and application of a lexicon-based classification. [13] is similar to the proposed work since it addresses the same context (Brazilian stock market) and has an unsupervised approach for sentiment analysis. However, it differs from ours, once we deal with two distinct datasets (tweets and news), two different classification approaches (supervised and unsupervised) and the analysis of pre-processing techniques.

3. METHODOLOGY

Sentiment analysis can be viewed as a classification problem with classes that represent the respective sentiment. The process for classifying text instances is well-known and has some main steps that are common to every classification task: data gathering, pre-processing, classifying and analyzing the results. In this work, these steps were followed, according to Figure 1, and are described next.

(a) **Data Gathering** Two different data sets regarding Brazilian stock market were used in the experiments. They contain 922 Twitter messages (tweets) and 373 news written in Brazilian Portuguese.

(b) **Pre-processing** All data sets were pre-processed to remove accents and punctuation marks, to have all words in lower case and to filter stopwords and numbers. Later, different pre-processing techniques were used to verify what was their real impact.

(c) **Classification** In this step, three different classifiers were used: a lexicon-based inspired by SOCAL, Naive-Bayes and Naive-Bayes multinomial. The objective of this paper was to compare supervised and unsupervised approaches and, as future work, find a way to combine both techniques. The Naive-Bayes classifiers were chosen because they are commonly used for text classification, have a simple model and yield reasonable precision and recall as shown in [17]. On the other hand, the SOCAL method is usually a reference regarding lexicon classifiers.

- SOCAL: Proposed by [21] consists of calculating a score for each instance, according to words semantic orientations (polarity and strength) found in a dictionary. It also accounts for negation, intensification and modal expressions. For Portuguese, there is an adaptation proposed by [15], which was applied in this work along with the SentiLex-PT polarity dictionary [18]. This method classifies instances in three classes: positive, negative or neutral.
- Naive-Bayes and Naive-Bayes Multinomial: Machine learning classifiers that according to the training set, construct a model based on probabilities of each word occurrence per class. For each test instance, an aggregated probability is calculated for each class and the one with higher probability labels the instance. In this step, the Weka tool [12] was used as the implementation of the classifiers. Usually, for sentiment analysis, the performance with three classes (positive, negative and neutral) is worse and the positive and negative classes are of greater interest. For this reason, a classification with two classes was chosen.

(d) **Result analysis** Three metrics were calculate: precision, recall and F-Measure, as defined in [14].

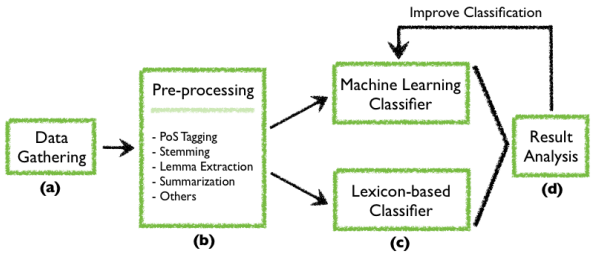


Figure 1: Methodology.

4. CASE STUDY AND RESULTS

4.1 Data sets

The experiments were performed with two data sets: Brazilian stock market news (SMN) and Brazilian stock market tweets (SMT), both gathered between August and December of 2013. They contain 373 and 922 instances, respectively, were written in Brazilian Portuguese and were manually labeled by a group of finance professionals. For SMN, instances have an average length of 202.10 words and of 17.21 hits in SentiLex-PT per document. Those news were extracted from 52 different sources. For SMT, instances have an average of 11.11 words/document and of 0.82 hits in SentiLex-PT per document. Regarding the users, there are 799 different users and an average of 1.15 tweet/user.

4.2 Pre-processing

As stated in Section 3, all instances of the data sets had accents and punctuation marks removed. For the machine learning classifiers, different pre-processing techniques were applied:

- Lemma extraction: all verbs are reduced to the infinitive form and nouns and adjectives to the singular male form. In this work, we used the tool developed at the NILC USP[1].
- Stemming: this tool reduces each word to its radical. For this, the Python library NLTK [4] was used.
- Part of Speech (PoS) Tagging: the NLPNET Python library [10], which offers a PoS tagger for structured texts, was used.
- Summarization: only the first three lines and the title of the news were considered for the classification step. This is consistent with extractive summarization approaches applied to Web.

4.3 Results

A ten-fold cross-validation was performed for Naive-Bayes classifiers and, for comparison, a random classifier was implemented with equal probabilities for each class. As the data sets have different properties, the results are presented in separate tables. In each row, the first string represents the classifier: Naive-Bayes (NB), Naive-Bayes Multinomial (NBM), SOCAL (SCL) and Random (RDM) and the subsequent words, represent the pre-processing techniques described in 4.2. There are two different ones that are *Title*, which consider only the titles of the news and *Lemma Verb*, which takes into account only the verbs of the instance in its lemma form. For the SCL method, there are different rows that represent the minimum of hits in the dictionary. The results are the weighted mean between the positive and the negative classes. The most relevant experiments, for the news data set (SMN), are reported in Table 1.

Data set	Precision	Recall	F-Measure
NB	0.747	0.74	0.742
NBM	**0.779**	**0.777**	**0.778**
NB Lemma	0.743	0.74	0.741
NBM Lemma	0.764	0.764	0.764
NB Stemming	0.736	0.733	0.734
NBM Stemming	0.751	0.753	0.752
NB Lemma Verb	0.669	0.669	0.669
NBM Lemma Verb	0.69	0.694	0.686
NB Title	0.699	0.702	0.696
NBM Title	0.736	0.724	0.726
NB Summarization	0.711	0.713	0.712
NBM Summarization	0.759	0.759	0.753
SCL 1-word	0.472	0.575	0.518
SCL 5-words	0.478	0.588	0.527
SCL 9-words	0.468	0.596	0.524
RDM	0.377	0.328	0.350

Table 1: Precision, Recall and F-Measure for data set SMN.

Table 1 shows many interesting results. Firstly, having more words in each instance contributes for a better performance of the classifiers, what is clear because summarization and only the title of the news presented worse results compared with the whole text. Another interesting result is that lemma extraction and stemming did not improve the precision or the recall of classifiers. This result could mean that the conjugation of words plays an important role in the classification method. It is also possible to verify that not only verbs, but other PoS are relevant to identify the sentiment of an instance, as the precision and recall were considerably worse when compared with the entire text. Ultimately, the machine-learning classifiers outperformed the lexicon-based approaches. Furthermore, comparing the results for SCL, there was not a significant difference between the results regarding the number of words found. Even though the performance of SCL was not as good as the NB and NBM, it is significantly better than the performance of the Random classifier.

Data set	Precision	Recall	F-Measure
NB	0.651	0.64	0.643
NBM	0.732	0.734	0.732
NB Lemma	0.67	0.663	0.665
NBM Lemma	**0.734**	**0.735**	**0.734**
NB Stemming	0.671	0.662	0.664
NBM Stemming	0.732	0.733	0.732
NB Lemma Verb	0.664	0.655	0.608
NBM Lemma Verb	0.7	0.702	0.689
SCL 1-word	0.522	0.518	0.520
SCL 3-words	0.672	0.676	0.674
RDM	0.369	0.340	0.354

Table 2: Precision, Recall and F-Measure for data set SMT.

Table 2 shows that every pre-processing technique applied to SMT improved NB results. On the other hand, only lemma extraction had a minor gain for the NBM. The improvements of pre-processing techniques could be explained by the fact that lemma extraction and stemming transform the instances in a manner that different words, usually with the same meaning, become the same string. Considering the SCL, it was significantly better than the Random classifier. Furthermore, the results for 3-words were better than some machine learning approaches. However, this is not highly significant because less than 10% of the data set have 3-words or more hits. Ultimately, the overall results for NB and NBM are worse then the ones of news dataset, which was expected as Twitter messages are highly unstructured (commonly, do not follow grammatical rules and word spelling). Another difference is the fact that for the SMN data set, the pre-processing techniques did not yield better precision or recall and, for the SMT, this was observed.

The tag cloud in Figure 2 represents the words and their relations with the classes. The green words are positive and the black words are negative, also the bigger and the thicker the word is, the more positive or negative it is.



Figure 2: Tag cloud.

It is possible to identify words that, as expected, are

[1]http://www.nilc.icmc.usp.br/nilc/index.php/toolsand-resources

clearly positive as *resultar*, *facilitar* and *aproximar* and others that are clearly negative as *desembolsar*, *rejeitar* and *despenca*.

5. CONCLUSION

Sentiment analysis techniques can be extremely useful in a great variety of contexts, including the understanding of social media streams. The experiments reported in this paper indicate that sentiment analysis can be successfully applied on Brazilian Portuguese stock market data, despite the scarcity of natural language processing tools. They also showed the impact of different pre-processing techniques and feature choices. Despite the very promising results (0.78 F-score), the techniques described in this paper can be further improved. In particular, investigating whether one can achieve even better results by combining lexicon and machine-learning approaches would be an interesting extension of this work.

Regarding the brazilian stock market data, it would be interesting as a future work trying to find correlation between the sentiment analysis results and stock market indexes such as operations volume.

Acknowledgment

This research was supported by the Brazilian National Institute of Science and Technology for the Web (CNPq grant numbers 573871/2008-6 and 477709/2012-5), CAPES, CNPq, Finep, Fapemig and Mitacs Globalink Program.

6. REFERENCES

[1] Seu futuro bem investido - relatório anual *bm&fbovespa* 2013. `http://ri.bmfbovespa.com.br/ptb/2002/RELATORIOWEBPORTUGUESFINAL.pdf`. Accessed: 2014-07-20.

[2] Ten largest stock exchanges in the world by market capitalization in 2011. `http://www.world-stock-exchanges.net/top10.html`. Accessed: 2014-07-20.

[3] World's largest stock exchanges. `http://www.relbanks.com/stock-exchanges/largest-stock-exchanges`. Accessed: 2014-07-20.

[4] S. Bird, E. Klein, and E. Loper. *Natural Language Processing with Python*. O'Reilly Media, Inc., 1st edition, 2009.

[5] J. Bollen, H. Mao, and X. Zeng. Twitter mood predicts the stock market. *Journal of Computational Science*, 2(1):1 – 8, 2011.

[6] W. F. M. D. Bondt and R. Thaler. Does the stock market overreact? *The Journal of Finance*, 40(3):793–805, 1986.

[7] N.-F. Chen, R. Roll, and S. A. Ross. Economic forces and the stock market. *The Journal of Business*, 59(3):83–403, 1986.

[8] E. F. Fama. The behavior of stock-market prices. *The Journal of Business*, 38(1):34–105, 1965.

[9] R. Feldman. Techniques and applications for sentiment analysis. *Commun. ACM*, 56(4):82–89, Apr. 2013.

[10] E. Fonseca and J. Rosa. A two-step convolutional neural network approach for semantic role labeling. In *Neural Networks (IJCNN), The 2013 International Joint Conference on*, pages 1–7, Aug 2013.

[11] S. Goel, J. M. Hofman, S. Lahaie, D. M. Pennock, and D. J. Watts. Predicting consumer behavior with web search. *Proceedings of the National Academy of Sciences*, 2010.

[12] M. Hall, E. Frank, G. Holmes, B. Pfahringer, P. Reutemann, and I. H. Witten. The weka data mining software: An update. *SIGKDD Explor. Newsl.*, 11(1):10–18, Nov. 2009.

[13] T. J. P. Lopes, G. K. L. Hiratani, F. J. Barth, O. Rodrigues, Jr., and J. M. Pinto. Mineração de opiniões aplicada à análise de investimentos. In *Companion Proceedings of the XIV Brazilian Symposium on Multimedia and the Web*, WebMedia '08, pages 117–120, Vila Velha, Espírito Santo, Brazil, 2008.

[14] C. D. Manning and H. Schütze. *Foundations of Statistical Natural Language Processing*. MIT Press, Cambridge, MA, USA, 1999.

[15] P. P. Balage Filho, T. A. S. Pardo, and S. M. Aluísio. An evaluation of the brazilian portuguese liwc dictionary for sentiment analysis. In *In 9th Brazilian Symposium in Information and Human Language Technology*, Fortaleza, Ceará, 2013.

[16] B. Pang and L. Lee. Opinion mining and sentiment analysis. *Found. Trends Inf. Retr.*, 2(1-2):1–135, Jan. 2008.

[17] B. Pang, L. Lee, and S. Vaithyanathan. Thumbs up?: Sentiment classification using machine learning techniques. In *Proceedings of the ACL-02 Conference on Empirical Methods in Natural Language Processing - Volume 10*, EMNLP '02, pages 79–86, Stroudsburg, PA, USA, 2002. Association for Computational Linguistics.

[18] M. J. Silva, P. Carvalho, and L. Sarmento. Building a sentiment lexicon for social judgement mining. In *Proceedings of the 10th International Conference on Computational Processing of the Portuguese Language*, PROPOR'12, pages 218–228, Berlin, Heidelberg, 2012. Springer-Verlag.

[19] M. J. Silva and R. TEAM. Notas sobre a realização e qualidade do twitómetro. Technical report, University of Lisbon, Faculty of Sciences, LASIGE, Lisbon, Portugal, May 2011.

[20] M. Souza and R. Vieira. Sentiment analysis on twitter data for portuguese language. In H. Caseli, A. Villavicencio, A. Teixeira, and F. Perdigão, editors, *Computational Processing of the Portuguese Language*, volume 7243 of *Lecture Notes in Computer Science*, pages 241–247. Springer Berlin Heidelberg, 2012.

[21] M. Taboada, J. Brooke, M. Tofiloski, K. Voll, and M. Stede. Lexicon-based methods for sentiment analysis. *Comput. Linguist.*, 37(2):267–307, June 2011.

[22] J. Yi, T. Nasukawa, R. Bunescu, and W. Niblack. Sentiment analyzer: Extracting sentiments about a given topic using natural language processing techniques. In *Proceedings of the Third IEEE International Conference on Data Mining*, ICDM '03, pages 427–, Washington, DC, USA, 2003. IEEE Computer Society.